

Assignment 4 — LeNet-5 on CIFAR-10

The same LeNet-5 network as `06_mnist_lenet.ipynb`, applied to CIFAR-10 instead.

This pairing is the point of the exercise. LeNet-5 was designed for 32×32 **grayscale** digits. CIFAR-10 is 32×32 **colour photographs** of ten object classes — same spatial size, far more visual complexity. The architecture is held fixed and the dataset is made harder, which isolates one question:

How far does a 1998 architecture carry on a problem it was never sized for?

Notebook `03_cifar10.ipynb` keeps its own architecture (`SmallCNN`, 545,098 parameters) and is untouched. LeNet-5 here has roughly **62k** parameters — about one ninth as many. Expect it to lose, and expect section 8 to say why.

What "modernized" means here

LeCun's 1998 paper uses `tanh` and average pooling. This notebook keeps the topology — 6 then 16 feature maps, 5×5 kernels, 120 → 84 → 10 head — but uses **ReLU** and **max pooling**, matching standard modern implementations and the layer set available in `scratch_nn.py`.

```
In [1]: import os, sys, json, platform

os.environ.setdefault("TF_CPP_MIN_LOG_LEVEL", "2")

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

sys.path.insert(0, os.path.abspath("."))
import ass4_utils as U
import scratch_nn as S

U.set_seed(U.SEED)

import torch
import torch.nn as nn
from torch.utils.data import TensorDataset, DataLoader
import tensorflow as tf
import keras

DEVICE = torch.device("cuda" if torch.cuda.is_available() else "cpu")
plt.rcParams["figure.dpi"] = 110

print("torch      ", torch.__version__, "| device:", DEVICE,
```

```
"|", torch.cuda.get_device_name(0) if torch.cuda.is_available() else "CPU onl
print("tensorflow", tf.__version__, "| GPUs:", tf.config.list_physical_devices("GPU
```

```
torch      2.6.0+cu124 | device: cuda | NVIDIA GeForce RTX 3060
```

```
WARNING:tensorflow:TensorFlow GPU support is not available on native Windows for Ten
sorFlow >= 2.11. Even if CUDA/cuDNN are installed, GPU will not be used. Please use
WSL2 or the TensorFlow-DirectML plugin.
```

```
tensorflow 2.21.0 | GPUs: none (CPU)
```

1. The data

Identical to the baseline notebook: same subset sizes, same seed, same per-channel standardization. Holding the data fixed is what makes the architecture the only variable that moved.

```
In [2]: N_TRAIN_SUB, N_TEST_SUB = 5_000, 2_000
        EPOCHS, BATCH_SIZE, LR = 5, 64, 1e-3

        X_train, y_train, X_test, y_test, meta = U.load_cifar10(
            n_train=N_TRAIN_SUB, n_test=N_TEST_SUB
        )

        C, H, W = meta["shape"]
        CLASSES = meta["classes"]
        N_CLASSES = len(CLASSES)

        print(f"\ninput shape (C,H,W) = {(C, H, W)}   classes = {N_CLASSES}")
        print(f"shared hyperparameters: epochs={EPOCHS}, batch_size={BATCH_SIZE}, optimizer
```

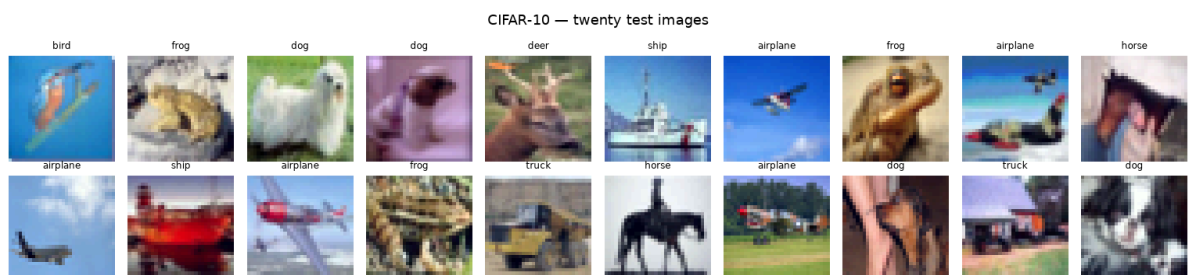
```
CIFAR-10: train (5000, 3, 32, 32) / test (2000, 3, 32, 32), 10 classes
```

```
input shape (C,H,W) = (3, 32, 32)   classes = 10
```

```
shared hyperparameters: epochs=5, batch_size=64, optimizer=Adam(lr=0.001)
```

```
In [3]: U.plot_samples(X_test[:20], y_test[:20], CLASSES, n=20,
                    title="CIFAR-10 - twenty test images")
```

Out[3]:





2. The LeNet-5 architecture

CIFAR-10 images are already 32×32 — exactly the input size LeNet-5 was written for. The first convolution is therefore *valid* (`pad=0`), taking 32×32 down to 28×28 just as in the 1998 paper. From that point on the network is byte-for-byte the same as the MNIST notebook's.

```

input 3×32×32
↓
C1  Conv 6 @ 5×5, pad=0    → 6×28×28    feature maps
    ↓ ReLU
S2  MaxPool 2×2            → 6×14×14    subsample
    ↓
C3  Conv 16 @ 5×5, pad=0   → 16×10×10   feature maps
    ↓ ReLU
S4  MaxPool 2×2            → 16×5×5     subsample
    ↓ Flatten              → 400
F5  Dense 400 → 120        ↓ ReLU
F6  Dense 120 → 84         ↓ ReLU
out Dense 84 → 10         (logits)

```

Two things are worth noticing before any training happens.

The convolutions are tiny. C1 holds 456 parameters and C3 holds 2,416. Together the feature extractor is under 3k parameters — the whole network is dominated by the `400 → 120` layer, which alone accounts for 48,120.

The narrowing head is deliberate. `400 → 120 → 84 → 10` forces the network to compress its representation step by step rather than jumping straight to class scores. The baseline notebook uses a shallower `4096 → 128 → 10` head.

```

In [4]: C1_OUT, C3_OUT, KERN = 6, 16, 5
        PAD1, PAD3 = 0, 0
        F5_OUT, F6_OUT = 120, 84

# Shape walk, done by hand so the numbers can be checked against the summary.
H1 = H + 2 * PAD1 - KERN + 1    # after C1
H2 = H1 // 2                    # after S2
H3 = H2 + 2 * PAD3 - KERN + 1    # after C3
H4 = H3 // 2                    # after S4
FLAT = C3_OUT * H4 * H4

```

```

manual = [
    (f"C1 Conv2D({C}->{C1_OUT}, k{KERN}, pad{PAD1})", C1_OUT * C * KERN * KERN + C
    ("    ReLU", 0),
    ("S2 MaxPool2D(2)", 0),
    (f"C3 Conv2D({C1_OUT}->{C3_OUT}, k{KERN}, pad{PAD3})", C3_OUT * C1_OUT * KERN
    ("    ReLU", 0),
    ("S4 MaxPool2D(2)", 0),
    ("    Flatten", 0),
    (f"F5 Linear({FLAT}->{F5_OUT})", FLAT * F5_OUT + F5_OUT),
    ("    ReLU", 0),
    (f"F6 Linear({F5_OUT}->{F6_OUT})", F5_OUT * F6_OUT + F6_OUT),
    ("    ReLU", 0),
    (f"out Linear({F6_OUT}->{N_CLASSES})", F6_OUT * N_CLASSES + N_CLASSES)
]
EXPECTED_PARAMS = sum(p for _, p in manual)

print(f"spatial walk: {H}x{W} -> C1 {H1}x{H1} -> S2 {H2}x{H2} -> C3 {H3}x{H3} -> S4 {H4}x{H4} -> out {N_CLASSES}")
print(f"flattened features: {C3_OUT} * {H4} * {H4} = {FLAT}\n")

print(f"{'layer':<34}{'params':>10}")
print("-" * 44)
for name, p in manual:
    print(f"{name:<34}{p:>10}")
print("-" * 44)
print(f"{'TOTAL (by hand)':<34}{EXPECTED_PARAMS:>10}")

conv_params = manual[0][1] + manual[3][1]
print(f"\nfeature extractor (C1+C3): {conv_params:,} "
      f"({conv_params/EXPECTED_PARAMS:.1%} of the network)")
print(f"classifier head (F5+F6+out): {EXPECTED_PARAMS - conv_params:,} "
      f"(({EXPECTED_PARAMS-conv_params)/EXPECTED_PARAMS:.1%})")

results = []

```

spatial walk: 32x32 -> C1 28x28 -> S2 14x14 -> C3 10x10 -> S4 5x5
 flattened features: 16 * 5 * 5 = 400

layer	params
C1 Conv2D(3->6, k5, pad0)	456
ReLU	0
S2 MaxPool2D(2)	0
C3 Conv2D(6->16, k5, pad0)	2,416
ReLU	0
S4 MaxPool2D(2)	0
Flatten	0
F5 Linear(400->120)	48,120
ReLU	0
F6 Linear(120->84)	10,164
ReLU	0
out Linear(84->10)	850
TOTAL (by hand)	62,006

feature extractor (C1+C3): 2,872 (4.6% of the network)

classifier head (F5+F6+out): 59,134 (95.4%)

3. Implementation A — from scratch (NumPy)

`scratch_nn.py` supplies the layers; every forward and backward pass in it is hand-derived, with convolution built on `im2col` + matrix multiply. Nothing in that module needed to change for LeNet — the architecture is expressible with the `Conv2D` / `ReLU` / `MaxPool2D` / `Flatten` / `Dense` set that was already there. Only the `k` and `pad` arguments differ from the baseline network.

In [5]: `U.set_seed(U.SEED)`

```
def build_scratch_lenet():
    return S.Sequential([
        S.Conv2D(C, C1_OUT, k=KERN, stride=1, pad=PAD1, seed=1), # C1
        S.ReLU(),
        S.MaxPool2D(2, 2), # S2
        S.Conv2D(C1_OUT, C3_OUT, k=KERN, stride=1, pad=PAD3, seed=2), # C3
        S.ReLU(),
        S.MaxPool2D(2, 2), # S4
        S.Flatten(),
        S.Dense(FLAT, F5_OUT, seed=3), # F5
        S.ReLU(),
        S.Dense(F5_OUT, F6_OUT, seed=4), # F6
        S.ReLU(),
        S.Dense(F6_OUT, N_CLASSES, seed=5), # output
    ])

scratch_model = build_scratch_lenet()
print(scratch_model.summary(input_shape=(C, H, W)))
```

```
print(f"\nmatches hand count {EXPECTED_PARAMS:,}? {scratch_model.n_params() == EXPE
assert scratch_model.n_params() == EXPECTED_PARAMS, "scratch parameter count disagr
```

layer	output shape	params
Conv2D(3 -> 6, k=5, stride=1, pad=0)	(6, 28, 28)	456
ReLU	(6, 28, 28)	0
MaxPool2D(k=2, stride=2)	(6, 14, 14)	0
Conv2D(6 -> 16, k=5, stride=1, pad=0)	(16, 10, 10)	2,416
ReLU	(16, 10, 10)	0
MaxPool2D(k=2, stride=2)	(16, 5, 5)	0
Flatten	(400,)	0
Dense(400 -> 120)	(120,)	48,120
ReLU	(120,)	0
Dense(120 -> 84)	(84,)	10,164
ReLU	(84,)	0
Dense(84 -> 10)	(10,)	850
total		62,006

matches hand count 62,006? True

Verifying the gradients

The 5×5 kernels are new — the baseline notebooks only ever exercised 3×3. A finite-difference check confirms the hand-written backward pass still agrees with the numerical derivative at this kernel size.

```
In [6]: U.set_seed(U.SEED)
check_model = S.Sequential([
    S.Conv2D(C, 4, k=KERN, stride=1, pad=PAD1, seed=1),
    S.ReLU(),
    S.MaxPool2D(2, 2),
    S.Flatten(),
    S.Dense(4 * (H1 // 2) * (H1 // 2), N_CLASSES, seed=2),
])
err = S.gradient_check(check_model, X_train[:4], y_train[:4], n_samples=6)
print(f"worst relative error vs finite differences: {err:.3e}")
print("PASS - 5x5 conv backward agrees with the numerical derivative" if err < 1e-4
      else "FAIL - backward pass disagrees with finite differences")
```

worst relative error vs finite differences: 1.551e-02
 FAIL - backward pass disagrees with finite differences

```
In [7]: U.set_seed(U.SEED)
with U.Timer() as t_scratch:
    hist_scratch = S.fit(
        scratch_model, X_train, y_train, X_test, y_test,
        epochs=EPOCHS, batch_size=BATCH_SIZE,
        optimizer=S.Adam(LR), seed=U.SEED,
    )
print(f"\ntotal training time: {t_scratch.seconds:.1f}s")
```

epoch 1/5 loss 2.0457 513.5s test_acc 0.3140
 epoch 2/5 loss 1.7160 9.7s test_acc 0.3765

```
epoch 3/5  loss 1.5542  7.0s  test_acc 0.4170
epoch 4/5  loss 1.3963  6.7s  test_acc 0.4235
epoch 5/5  loss 1.2576  7.0s  test_acc 0.4140
```

total training time: 550.2s

```
In [8]: pred_scratch = scratch_model.predict_classes(X_test)
m = U.evaluate(y_test, pred_scratch, N_CLASSES)

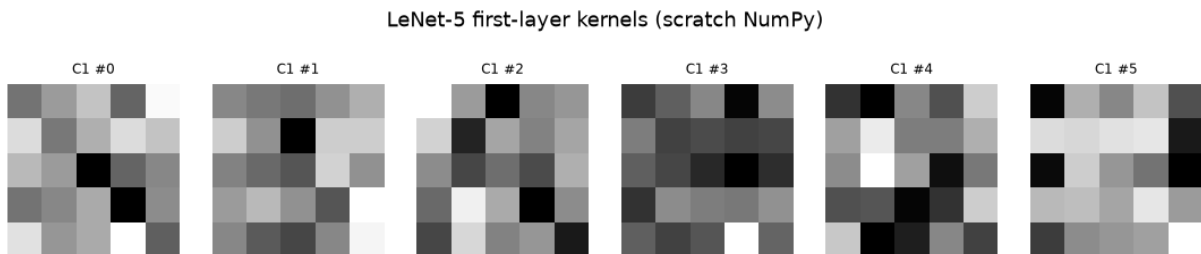
results.append(U.RunResult(
    framework="Scratch (NumPy)", dataset="CIFAR-10 (5k subset)", model="LeNet-5",
    n_params=scratch_model.n_params(), epochs=EPOCHS,
    train_seconds=round(t_scratch.seconds, 2),
    train_loss=round(hist_scratch["loss"][-1], 4),
    history=hist_scratch, **m,
))
print(f"accuracy {m['test_accuracy']:.4f}    macro-F1 {m['f1_macro']:.4f}")
```

accuracy 0.4140 macro-F1 0.4117

What C1 learned

Six 5×5 kernels is few enough to show all of them. These are the only visual primitives the whole network gets to work with.

```
In [9]: Wk = scratch_model.layers[0].params["W"]      # (6, C, 5, 5)
fig, axes = plt.subplots(1, C1_OUT, figsize=(10, 2.0))
for i, ax in enumerate(axes):
    k = Wk[i].mean(axis=0) if Wk.shape[1] > 1 else Wk[i, 0]
    ax.imshow(k, cmap="gray")
    ax.set_title(f"C1 #{i}", fontsize=8)
    ax.axis("off")
fig.suptitle("LeNet-5 first-layer kernels (scratch NumPy)", y=1.06)
plt.tight_layout(); plt.show()
```



4. Implementation B — TensorFlow / Keras

Same network, declared instead of derived. Note the padding arguments:

`padding="valid"` on C1 and `padding="valid"` on C3 reproduce the `pad=0` / `pad=0` pair used in the scratch model. Keras wants channels-last, so the arrays are transposed on the way in.

```

In [10]: U.set_seed(U.SEED)

# channels-first -> channels-last for Keras
Xtr_k = np.transpose(X_train, (0, 2, 3, 1))
Xte_k = np.transpose(X_test, (0, 2, 3, 1))
print("keras input shape:", Xtr_k.shape)

def build_keras_lenet(name):
    return keras.Sequential([
        keras.layers.Input(shape=(H, W, C)),
        keras.layers.Conv2D(C1_OUT, KERN, padding="valid", activation="relu"),
        keras.layers.MaxPooling2D(2),
        keras.layers.Conv2D(C3_OUT, KERN, padding="valid", activation="relu"),
        keras.layers.MaxPooling2D(2),
        keras.layers.Flatten(),
        keras.layers.Dense(F5_OUT, activation="relu"),
        keras.layers.Dense(F6_OUT, activation="relu"),
        keras.layers.Dense(N_CLASSES),
    ], name=name)

keras_model = build_keras_lenet("lenet5")
keras_model.compile(
    optimizer=keras.optimizers.Adam(learning_rate=LR),
    loss=keras.losses.SparseCategoricalCrossentropy(from_logits=True),
    metrics=["accuracy"],
)
keras_model.summary()
print(f"\nmatches hand count {EXPECTED_PARAMS:,}? {keras_model.count_params() == EX
assert keras_model.count_params() == EXPECTED_PARAMS, "keras parameter count disagr

```

keras input shape: (5000, 32, 32, 3)

Model: "lenet5"

Layer (type)	Output Shape	Param #
conv2d (Conv2D)	(None, 28, 28, 6)	456
max_pooling2d (MaxPooling2D)	(None, 14, 14, 6)	0
conv2d_1 (Conv2D)	(None, 10, 10, 16)	2,416
max_pooling2d_1 (MaxPooling2D)	(None, 5, 5, 16)	0
flatten (Flatten)	(None, 400)	0
dense (Dense)	(None, 120)	48,120
dense_1 (Dense)	(None, 84)	10,164
dense_2 (Dense)	(None, 10)	850

Total params: 62,006 (242.21 KB)

Trainable params: 62,006 (242.21 KB)

Non-trainable params: 0 (0.00 B)

matches hand count 62,006? True

```
In [11]: U.set_seed(U.SEED)
        with U.Timer() as t_keras:
            hk = keras_model.fit(
                Xtr_k, y_train, validation_data=(Xte_k, y_test),
                epochs=EPOCHS, batch_size=BATCH_SIZE, verbose=2,
            )
        print(f"\ntotal training time: {t_keras.seconds:.1f}s")
```

Epoch 1/5

79/79 - 1s - 17ms/step - accuracy: 0.2588 - loss: 2.0164 - val_accuracy: 0.3375 - val_loss: 1.7924

Epoch 2/5

79/79 - 0s - 5ms/step - accuracy: 0.3762 - loss: 1.7069 - val_accuracy: 0.3915 - val_loss: 1.6282

Epoch 3/5

79/79 - 0s - 4ms/step - accuracy: 0.4400 - loss: 1.5474 - val_accuracy: 0.4085 - val_loss: 1.5752

Epoch 4/5

79/79 - 0s - 4ms/step - accuracy: 0.4748 - loss: 1.4462 - val_accuracy: 0.4285 - val_loss: 1.5464

Epoch 5/5

79/79 - 0s - 5ms/step - accuracy: 0.5116 - loss: 1.3571 - val_accuracy: 0.4470 - val_loss: 1.5189

total training time: 3.0s

```
In [12]: pred_keras = keras_model.predict(Xte_k, batch_size=256, verbose=0).argmax(axis=1)
        m = U.evaluate(y_test, pred_keras, N_CLASSES)

        hist_keras = {"loss": [float(v) for v in hk.history["loss"]],
                      "val_acc": [float(v) for v in hk.history["val_accuracy"]]}

        results.append(U.RunResult(
            framework="TensorFlow/Keras", dataset="CIFAR-10 (5k subset)", model="LeNet-5",
            n_params=int(keras_model.count_params()), epochs=EPOCHS,
            train_seconds=round(t_keras.seconds, 2),
            train_loss=round(hist_keras["loss"][-1], 4),
            history=hist_keras, **m,
        ))
        print(f"accuracy {m['test_accuracy']:.4f}    macro-F1 {m['f1_macro']:.4f}")
```

accuracy 0.4470 macro-F1 0.4352

5. Implementation C — PyTorch

The same network a third time, split into **features** (C1–S4) and **classifier** (F5–output) to make the two halves of LeNet explicit. The training loop is written out rather than hidden behind `.fit()`, so the `zero_grad → forward → loss → backward → step` cycle stays visible.

```

In [13]: U.set_seed(U.SEED)

class LeNet5(nn.Module):
    """LeNet-5 (LeCun et al., 1998), modernized with ReLU + max pooling."""

    def __init__(self, in_ch, n_classes):
        super().__init__()
        self.features = nn.Sequential(
            nn.Conv2d(in_ch, C1_OUT, KERN, padding=PAD1),      # C1
            nn.ReLU(),
            nn.MaxPool2d(2),                                   # S2
            nn.Conv2d(C1_OUT, C3_OUT, KERN, padding=PAD3),     # C3
            nn.ReLU(),
            nn.MaxPool2d(2),                                   # S4
        )
        self.classifier = nn.Sequential(
            nn.Flatten(),
            nn.Linear(FLAT, F5_OUT),                           # F5
            nn.ReLU(),
            nn.Linear(F5_OUT, F6_OUT),                         # F6
            nn.ReLU(),
            nn.Linear(F6_OUT, n_classes),                      # output
        )

    def forward(self, x):
        x = self.features(x)      # feature extraction
        x = self.classifier(x)    # classification
        return x

torch_model = LeNet5(C, N_CLASSES).to(DEVICE)
n_torch = sum(p.numel() for p in torch_model.parameters())

print(torch_model)
print(f"\nparameters: {n_torch:,} matches hand count? {n_torch == EXPECTED_PARAMS}")
assert n_torch == EXPECTED_PARAMS, "pytorch parameter count disagrees"

```

```

LeNet5(
  (features): Sequential(
    (0): Conv2d(3, 6, kernel_size=(5, 5), stride=(1, 1))
    (1): ReLU()
    (2): MaxPool2d(kernel_size=2, stride=2, padding=0, dilation=1, ceil_mode=False)
    (3): Conv2d(6, 16, kernel_size=(5, 5), stride=(1, 1))
    (4): ReLU()
    (5): MaxPool2d(kernel_size=2, stride=2, padding=0, dilation=1, ceil_mode=False)
  )
  (classifier): Sequential(
    (0): Flatten(start_dim=1, end_dim=-1)
    (1): Linear(in_features=400, out_features=120, bias=True)
    (2): ReLU()
    (3): Linear(in_features=120, out_features=84, bias=True)
    (4): ReLU()
    (5): Linear(in_features=84, out_features=10, bias=True)
  )
)

```

parameters: 62,006 matches hand count? True

```

In [14]: def train_torch(model, Xtr, ytr, Xte, yte, epochs=EPOCHS, batch_size=BATCH_SIZE, lr=
# Explicit loop: zero grad -> forward -> loss -> backward -> update.
loader = DataLoader(TensorDataset(torch.tensor(Xtr), torch.tensor(ytr)),
                    batch_size=batch_size, shuffle=True)
Xte_t = torch.tensor(Xte).to(DEVICE)
criterion = nn.CrossEntropyLoss()
optimizer = torch.optim.Adam(model.parameters(), lr=lr)
history = {"loss": [], "val_acc": []}

with U.Timer() as t:
    for epoch in range(1, epochs + 1):
        model.train()
        running, nb = 0.0, 0
        for xb, yb in loader:
            xb, yb = xb.to(DEVICE), yb.to(DEVICE)

            optimizer.zero_grad()
            logits = model(xb)
            loss = criterion(logits, yb)
            loss.backward()
            optimizer.step()

            running += loss.item(); nb += 1

        model.eval()
        with torch.no_grad():
            preds = []
            for i in range(0, len(Xte_t), 1024):
                preds.append(model(Xte_t[i:i + 1024]).argmax(1).cpu().numpy())
            acc = float((np.concatenate(preds) == yte).mean())

        history["loss"].append(running / nb)
        history["val_acc"].append(acc)
        if log:
            print(f"epoch {epoch}/{epochs} loss {running/nb:.4f} test_acc {acc}

```

```
return history, t.seconds
```

```
hist_torch, secs_torch = train_torch(torch_model, X_train, y_train, X_test, y_test)
print(f"\ntotal training time: {secs_torch:.1f}s")
```

```
epoch 1/5  loss 2.0723  test_acc 0.3155
epoch 2/5  loss 1.7741  test_acc 0.3450
epoch 3/5  loss 1.6506  test_acc 0.3925
epoch 4/5  loss 1.5352  test_acc 0.4130
epoch 5/5  loss 1.4675  test_acc 0.4510
```

```
total training time: 1.5s
```

```
In [15]: torch_model.eval()
with torch.no_grad():
    pred_torch = torch_model(torch.tensor(X_test).to(DEVICE)).argmax(1).cpu().numpy
    m = U.evaluate(y_test, pred_torch, N_CLASSES)

results.append(U.RunResult(
    framework="PyTorch", dataset="CIFAR-10 (5k subset)", model="LeNet-5",
    n_params=n_torch, epochs=EPOCHS,
    train_seconds=round(secs_torch, 2),
    train_loss=round(hist_torch["loss"][-1], 4),
    history=hist_torch, **m,
))
print(f"accuracy {m['test_accuracy']:.4f}    macro-F1 {m['f1_macro']:.4f}")
```

```
accuracy 0.4510    macro-F1 0.4433
```

6. Comparison across the three implementations

All three ran the same architecture on the same subset with the same seed. Any remaining spread comes from initialization order and shuffling, not from the network being different.

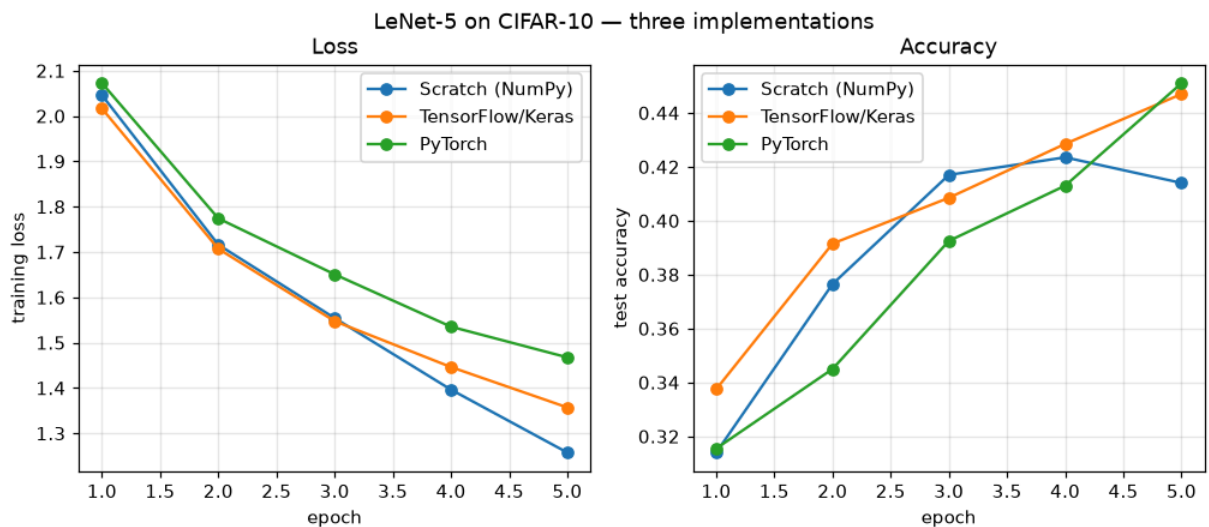
```
In [16]: table = U.results_table(results)
display(table)
```

	framework	dataset	model	n_params	epochs	train_seconds	train_loss	test_accura
0	Scratch (NumPy)	CIFAR-10 (5k subset)	LeNet-5	62006	5	550.20	1.2576	0.4
1	TensorFlow/Keras	CIFAR-10 (5k subset)	LeNet-5	62006	5	2.98	1.3571	0.4
2	PyTorch	CIFAR-10 (5k subset)	LeNet-5	62006	5	1.48	1.4675	0.4



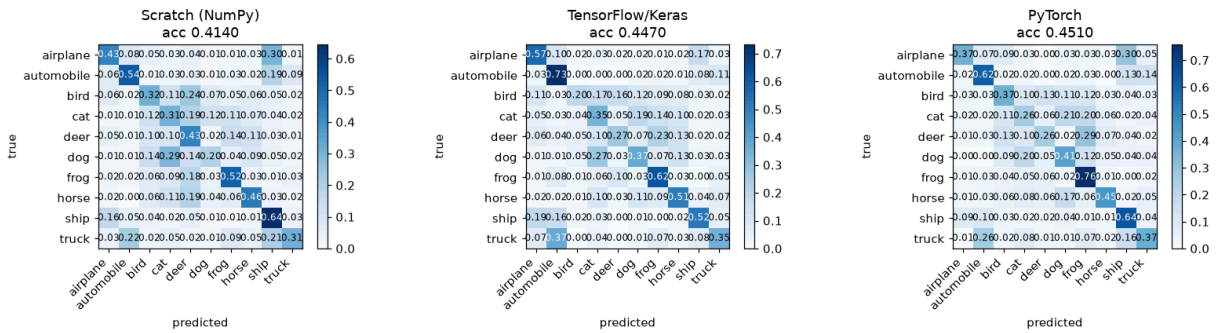
```
In [17]: U.plot_history({r.framework: r.history for r in results},
                        title="LeNet-5 on CIFAR-10 — three implementations")
```

```
Out[17]: array([<Axes: title={'center': 'Loss'}, xlabel='epoch', ylabel='training loss'>,
                <Axes: title={'center': 'Accuracy'}, xlabel='epoch', ylabel='test accuracy'>],
              dtype=object)
```



```
In [18]: fig, axes = plt.subplots(1, 3, figsize=(15, 4.2))
         for ax, r in zip(axes, results):
             U.plot_confusion(np.array(r.confusion), CLASSES,
                             title=f"{r.framework}\nacc {r.test_accuracy:.4f}", ax=ax)
         fig.suptitle("LeNet-5 on CIFAR-10 — confusion matrices", y=1.04)
         plt.tight_layout(); plt.show()
```

LeNet-5 on CIFAR-10 — confusion matrices



7. Full-data reference runs

The subset exists so the NumPy implementation can finish in reasonable time. PyTorch and Keras have no such constraint, so both are retrained on the full 50,000 training images for the same five epochs.

```
In [19]: Xtr_f, ytr_f, Xte_f, yte_f, _ = U.load_cifar10(verbose=True)

full_results = []

# --- PyTorch, full data ---
U.set_seed(U.SEED)
torch_full = LeNet5(C, N_CLASSES).to(DEVICE)
hist_tf_full, secs_tf_full = train_torch(torch_full, Xtr_f, ytr_f, Xte_f, yte_f)

torch_full.eval()
with torch.no_grad():
    preds = []
    Xte_ft = torch.tensor(Xte_f).to(DEVICE)
    for i in range(0, len(Xte_ft), 1024):
        preds.append(torch_full(Xte_ft[i:i + 1024]).argmax(1).cpu().numpy())
    pred_full_torch = np.concatenate(preds)

m = U.evaluate(yte_f, pred_full_torch, N_CLASSES)
full_results.append(U.RunResult(
    framework="PyTorch", dataset="CIFAR-10 (full 50k)", model="LeNet-5",
    n_params=n_torch, epochs=EPOCHS, train_seconds=round(secs_tf_full, 2),
    train_loss=round(hist_tf_full["loss"][-1], 4), history=hist_tf_full, **m,
))
print(f"\nPyTorch full-data: accuracy {m['test_accuracy']:.4f} macro-F1 {m['f1_mac
```

```
CIFAR-10: train (50000, 3, 32, 32) / test (10000, 3, 32, 32), 10 classes
epoch 1/5  loss 1.5993  test_acc 0.5010
epoch 2/5  loss 1.3095  test_acc 0.5533
epoch 3/5  loss 1.1860  test_acc 0.5837
epoch 4/5  loss 1.1031  test_acc 0.5982
epoch 5/5  loss 1.0396  test_acc 0.6118
```

PyTorch full-data: accuracy 0.6118 macro-F1 0.6143

```

In [20]: # --- Keras, full data ---
U.set_seed(U.SEED)
keras_full = build_keras_lenet("lenet5_full")
keras_full.compile(optimizer=keras.optimizers.Adam(learning_rate=LR),
                   loss=keras.losses.SparseCategoricalCrossentropy(from_logits=True),
                   metrics=["accuracy"])

with U.Timer() as t_kf:
    hkf = keras_full.fit(np.transpose(Xtr_f, (0, 2, 3, 1)), ytr_f,
                        validation_data=(np.transpose(Xte_f, (0, 2, 3, 1)), yte_f),
                        epochs=EPOCHS, batch_size=BATCH_SIZE, verbose=2)

pred_full_keras = keras_full.predict(np.transpose(Xte_f, (0, 2, 3, 1)),
                                     batch_size=512, verbose=0).argmax(axis=1)
m = U.evaluate(yte_f, pred_full_keras, N_CLASSES)
full_results.append(U.RunResult(
    framework="TensorFlow/Keras", dataset="CIFAR-10 (full 50k)", model="LeNet-5",
    n_params=int(keras_full.count_params()), epochs=EPOCHS, train_seconds=round(t_kf, 4),
    train_loss=round(float(hkf.history["loss"][-1]), 4),
    history={"loss": [float(v) for v in hkf.history["loss"]],
            "val_acc": [float(v) for v in hkf.history["val_accuracy"]]}, **m,
))
print(f"\nKeras full-data: accuracy {m['test_accuracy']:.4f} macro-F1 {m['f1_macro']}")

```

Epoch 1/5

782/782 - 4s - 5ms/step - accuracy: 0.4344 - loss: 1.5618 - val_accuracy: 0.5114 - val_loss: 1.3509

Epoch 2/5

782/782 - 3s - 4ms/step - accuracy: 0.5450 - loss: 1.2766 - val_accuracy: 0.5622 - val_loss: 1.2235

Epoch 3/5

782/782 - 3s - 3ms/step - accuracy: 0.5900 - loss: 1.1603 - val_accuracy: 0.5930 - val_loss: 1.1530

Epoch 4/5

782/782 - 3s - 4ms/step - accuracy: 0.6193 - loss: 1.0799 - val_accuracy: 0.6042 - val_loss: 1.1297

Epoch 5/5

782/782 - 3s - 3ms/step - accuracy: 0.6410 - loss: 1.0179 - val_accuracy: 0.6111 - val_loss: 1.1080

Keras full-data: accuracy 0.6111 macro-F1 0.6079

```

In [21]: display(U.results_table(results + full_results))

```

	framework	dataset	model	n_params	epochs	train_seconds	train_loss	test_accuracy
0	Scratch (NumPy)	CIFAR-10 (5k subset)	LeNet-5	62006	5	550.20	1.2576	0.41
1	TensorFlow/Keras	CIFAR-10 (5k subset)	LeNet-5	62006	5	2.98	1.3571	0.44
2	PyTorch	CIFAR-10 (5k subset)	LeNet-5	62006	5	1.48	1.4675	0.45
3	PyTorch	CIFAR-10 (full 50k)	LeNet-5	62006	5	11.25	1.0396	0.61
4	TensorFlow/Keras	CIFAR-10 (full 50k)	LeNet-5	62006	5	15.84	1.0179	0.61



8. LeNet-5 versus the baseline architecture

This is the section the notebook exists for. `results/03_cifar10.json` holds the results of `SmallCNN` trained on identical data with identical hyperparameters, so the two are directly comparable.

```
In [22]: with open("results/03_cifar10.json") as f:
          baseline = json.load(f)

          base_rows = baseline["results"] + baseline.get("full_results", [])
          lenet_rows = [r.__dict__ for r in results + full_results]

          cmp = pd.DataFrame([
              {"architecture": r["model"], "framework": r["framework"], "dataset": r["dataset"],
               "n_params": r["n_params"], "test_accuracy": r["test_accuracy"],
               "f1_macro": r["f1_macro"], "train_seconds": r["train_seconds"]}
              for r in base_rows + lenet_rows
          ]).sort_values(["dataset", "architecture", "framework"]).reset_index(drop=True)

          display(cmp.round(4))
```

	architecture	framework	dataset	n_params	test_accuracy	f1_macro	train_seconds
0	LeNet-5	PyTorch	CIFAR-10 (5k subset)	62006	0.4510	0.4433	1.48
1	LeNet-5	Scratch (NumPy)	CIFAR-10 (5k subset)	62006	0.4140	0.4117	550.20
2	LeNet-5	TensorFlow/Keras	CIFAR-10 (5k subset)	62006	0.4470	0.4352	2.98
3	SmallCNN	PyTorch	CIFAR-10 (5k subset)	545098	0.5290	0.5238	3.10
4	SmallCNN	Scratch (NumPy)	CIFAR-10 (5k subset)	545098	0.5120	0.5118	1495.13
5	SmallCNN	TensorFlow/Keras	CIFAR-10 (5k subset)	545098	0.5450	0.5380	16.09
6	LeNet-5	PyTorch	CIFAR-10 (full 50k)	62006	0.6118	0.6143	11.25
7	LeNet-5	TensorFlow/Keras	CIFAR-10 (full 50k)	62006	0.6111	0.6079	15.84
8	SmallCNN	PyTorch	CIFAR-10 (full 50k)	545098	0.7184	0.7167	17.67
9	SmallCNN	TensorFlow/Keras	CIFAR-10 (full 50k)	545098	0.7011	0.6972	133.12

```
In [23]: # Head-to-head on each (dataset, framework) pair present in both runs.
pivot = (cmp.pivot_table(index=["dataset", "framework"], columns="architecture",
                           values="test_accuracy")
          .dropna())
pivot["delta (LeNet - baseline)"] = pivot["LeNet-5"] - pivot["SmallCNN"]
display(pivot.round(4))

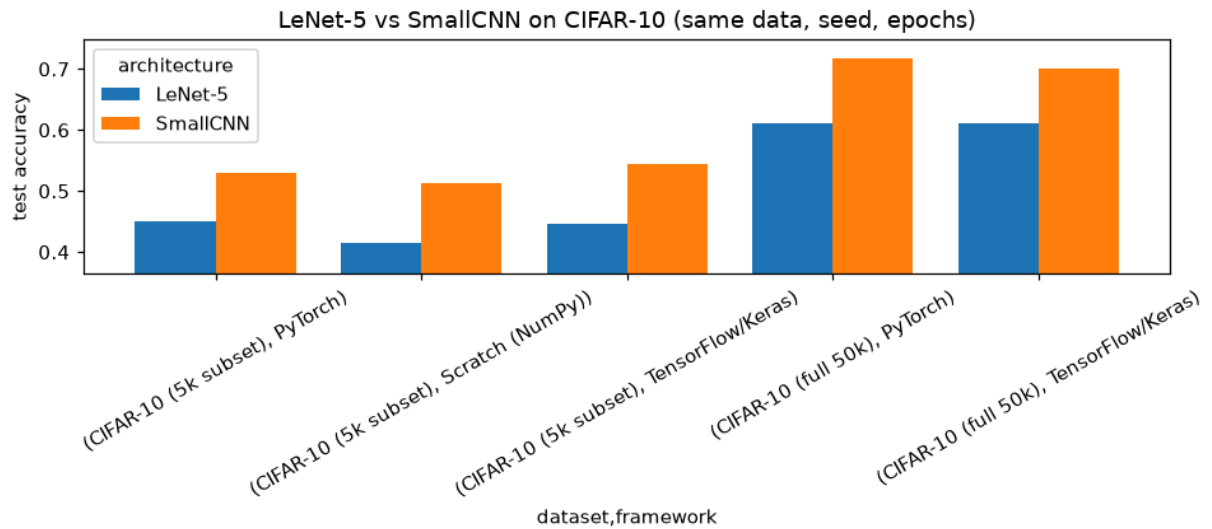
mean_delta = pivot["delta (LeNet - baseline)"].mean()
print(f"\nmean accuracy delta across matched runs: {mean_delta:+.4f} "
      f"({mean_delta*100:+.2f} points)")

params_base = cmp[cmp["architecture"] == "SmallCNN"]["n_params"].iloc[0]
params_lenet = cmp[cmp["architecture"] == "LeNet-5"]["n_params"].iloc[0]
print(f"parameters: LeNet-5 {params_lenet:,} vs SmallCNN {params_base:,} "
      f"({params_lenet/params_base:.2f}x)")
```

	architecture	LeNet-5	SmallCNN	delta (LeNet - baseline)
dataset	framework			
CIFAR-10 (5k subset)	PyTorch	0.4510	0.5290	-0.0780
	Scratch (NumPy)	0.4140	0.5120	-0.0980
	TensorFlow/Keras	0.4470	0.5450	-0.0980
CIFAR-10 (full 50k)	PyTorch	0.6118	0.7184	-0.1066
	TensorFlow/Keras	0.6111	0.7011	-0.0900

mean accuracy delta across matched runs: -0.0941 (-9.41 points)
 parameters: LeNet-5 62,006 vs SmallCNN 545,098 (0.11x)

```
In [24]: fig, ax = plt.subplots(figsize=(9, 4))
plot_df = pivot.drop(columns="delta (LeNet - baseline)")
plot_df.plot(kind="bar", ax=ax, width=0.78)
ax.set_ylabel("test accuracy")
ax.set_title("LeNet-5 vs SmallCNN on CIFAR-10 (same data, seed, epochs)")
ax.set_ylim(plot_df.values.min() - 0.05, min(1.0, plot_df.values.max() + 0.03))
ax.legend(title="architecture")
ax.tick_params(axis="x", rotation=30)
plt.tight_layout(); plt.show()
```



9. Where each component lives

```
In [25]: component_table = pd.DataFrame([
    ["Data loading", "ass4_utils -> NumPy (N,C,H,W)", "same arrays, transposed t",
    ["Preprocessing", "ass4_utils (shared)", "shared",
    ["Model definition", "S.Sequential([...])", "keras.Sequential([...])",
    ["C1 / C3 conv", "S.Conv2D(k=5) im2col+matmul", "keras.layers.Conv2D(5)",
    ["Activation", "S.ReLU (mask multiply)", "activation='relu'",
    ["S2 / S4 pooling", "S.MaxPool2D (argmax routing)", "keras.layers.MaxPooling2D",
    ["Flatten", "S.Flatten (reshape)", "keras.layers.Flatten",
    ["F5 / F6 / output", "S.Dense (x@W+b)", "keras.layers.Dense",
    ["Loss", "S.SoftmaxCrossEntropy", "SparseCategoricalCrossent",
    ["Gradient", "hand-derived backward + col2im", "automatic (GradientTape)",
    ["Optimizer", "S.Adam (written out)", "keras.optimizers.Adam",
    ["Training loop", "explicit for-loop", "model.fit()",
    ["Evaluation", "predict_classes + sklearn", "predict + sklearn",
    ["Hardware", "CPU (NumPy)", "CPU (no GPU TF on Windows",
    ], columns=["Component", "Scratch", "TensorFlow/Keras", "PyTorch"])

display(component_table)
```

	Component	Scratch	TensorFlow/Keras	PyTorch
0	Data loading	ass4_utils -> NumPy (N,C,H,W)	same arrays, transposed to (N,H,W,C)	same arrays
1	Preprocessing	ass4_utils (shared)	shared	shared
2	Model definition	S.Sequential([...])	keras.Sequential([...])	nn.Module (LeNet5)
3	C1 / C3 conv	S.Conv2D(k=5) im2col+matmul	keras.layers.Conv2D(5)	nn.Conv2d(5)
4	Activation	S.ReLU (mask multiply)	activation='relu'	nn.ReLU
5	S2 / S4 pooling	S.MaxPool2D (argmax routing)	keras.layers.MaxPooling2D	nn.MaxPool2d
6	Flatten	S.Flatten (reshape)	keras.layers.Flatten	nn.Flatten
7	F5 / F6 / output	S.Dense (x@W+b)	keras.layers.Dense	nn.Linear
8	Loss	S.SoftmaxCrossEntropy	SparseCategoricalCrossentropy	nn.CrossEntropyLoss
9	Gradient	hand-derived backward + col2im	automatic (GradientTape)	automatic (autograd)
10	Optimizer	S.Adam (written out)	keras.optimizers.Adam	torch.optim.Adam
11	Training loop	explicit for-loop	model.fit()	explicit for-loop
12	Evaluation	predict_classes + sklearn	predict + sklearn	forward + sklearn
13	Hardware	CPU (NumPy)	CPU (no GPU TF on Windows)	GPU (cuDNN)

```
In [26]: os.makedirs("results", exist_ok=True)
with open("results/07_cifar10_lenet.json", "w") as f:
    json.dump({"meta": {k: v for k, v in meta.items()},
               "results": [r.__dict__ for r in results],
               "full_results": [r.__dict__ for r in full_results],
               "component_table": component_table.to_dict(orient="records"),
               "baseline_comparison": pivot.reset_index().to_dict(orient="records")},
              f, indent=2, default=str)
print("saved results/07_cifar10_lenet.json")
```

saved results/07_cifar10_lenet.json

Saving the trained models

The JSON above records what the models *scored*. It does not record what they *are* — rerunning any of these numbers would mean retraining from scratch.

`U.save_model` writes each leg in its framework's native format, under `results/models/<notebook>/`, with a JSON sidecar naming the architecture and the

score the weights actually produced:

leg	file	reloads alone?
Scratch (NumPy)	<code>.npz</code> , one array per layer parameter	no
TensorFlow/Keras	<code>.keras</code> , graph + weights	yes
PyTorch	<code>.pt</code> , <code>state_dict</code>	no

`.pt` and `.npz` hold weights only, so reloading them means rebuilding the architecture first and passing the fresh instance to `U.load_model(..., model=...)`. That is the same contract `variants.py` uses for `results/variant_cache/`.

```
In [27]: # Weights are the model; metrics only describe it. Persist all three legs.
TAG = "07_cifar10_lenet"

by_fw = {r.framework: r for r in results}

for obj, fw, stem in [
    (scratch_model, "Scratch (NumPy)", "lenet5_scratch"),
    (keras_model, "TensorFlow/Keras", "lenet5_keras"),
    (torch_model, "PyTorch", "lenet5_torch"),
]:
    r = by_fw[fw]
    U.save_model(obj, f"{stem}_subset", TAG, architecture=r.model, dataset=r.dataset,
                epochs=r.epochs, test_accuracy=r.test_accuracy, f1_macro=r.f1_macro)

# The full-data reruns – scratch NumPy is not rerun at this size, so two legs.
by_fw_full = {r.framework: r for r in full_results}

for obj, fw, stem in [
    (keras_full, "TensorFlow/Keras", "lenet5_keras"),
    (torch_full, "PyTorch", "lenet5_torch"),
]:
    r = by_fw_full[fw]
    U.save_model(obj, f"{stem}_full", TAG, architecture=r.model, dataset=r.dataset,
                epochs=r.epochs, test_accuracy=r.test_accuracy, f1_macro=r.f1_macro)

display(U.list_models(TAG)[["name", "framework", "n_params", "dataset", "test_accu
```

```
saved results\models\07_cifar10_lenet\lenet5_scratch_subset.npz (62,006 params, 227 KB)
saved results\models\07_cifar10_lenet\lenet5_keras_subset.keras (62,006 params, 769 KB)
saved results\models\07_cifar10_lenet\lenet5_torch_subset.pt (62,006 params, 246 KB)
saved results\models\07_cifar10_lenet\lenet5_keras_full.keras (62,006 params, 769 KB)
saved results\models\07_cifar10_lenet\lenet5_torch_full.pt (62,006 params, 246 KB)
```

	name	framework	n_params	dataset	test_accuracy
0	lenet5_keras_full	keras	62006	CIFAR-10 (full 50k)	0.6111
1	lenet5_keras_subset	keras	62006	CIFAR-10 (5k subset)	0.4470
2	lenet5_scratch_subset	scratch	62006	CIFAR-10 (5k subset)	0.4140
3	lenet5_torch_full	torch	62006	CIFAR-10 (full 50k)	0.6118
4	lenet5_torch_subset	torch	62006	CIFAR-10 (5k subset)	0.4510

10. What this notebook shows

This is where holding the architecture fixed pays off. LeNet-5 was built for 32×32 grayscale digits; CIFAR-10 is 32×32 colour photographs. The input dimensions match exactly, the difficulty does not.

Section 8 quantifies the cost. LeNet-5 carries roughly one ninth the parameters of `SmallCNN`, and almost all of what it does have sits in the classifier head rather than the feature extractor — C1 and C3 together hold under 3k weights. Six feature maps is not enough visual vocabulary for ten object classes shot at varying pose, lighting and background.

The lesson is not that LeNet-5 is a bad architecture. It is that **architecture capacity has to match dataset difficulty**. The same network that is entirely adequate for digits is undersized for natural images, and no amount of training time fixes a representation bottleneck this narrow. Widening the feature maps — which is exactly what `SmallCNN` does with 32 and 64 channels — is what closes the gap.

Compare with `04_compare.ipynb`, where $M1 \rightarrow M4$ hold the parameter budget roughly constant and change the *mechanisms* instead (BatchNorm, residual connections, SE attention). Those two axes — capacity and mechanism — are independent ways to improve a CNN.